Samitha Athurupana

Full Stack Software Engineer | Python • Java • React • AWS • AI/ML

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PROFESSIONAL SUMMARY

Full Stack Software Engineer with 2+ years of experience designing and delivering end-to-end web applications using React, Python/FastAPI, and Java/Spring Boot. Proven expertise in RESTful API design, AWS cloud deployments, Docker, and Kubernetes. Hands-on experience building production AI/ML solutions including RAG pipelines, LangGraph autonomous agents, and LLM-powered applications. Currently pursuing an MSc in Information Technology while contributing to enterprise ERP and agriculture-tech platforms.

PROFESSIONAL EXPERIENCE

Associate Software Engineer | *Insight Square (Pvt) Ltd* | Horana, Sri Lanka **Jan 2024 – Present**

- Designed and delivered full-stack features across React frontends and Python/FastAPI & Java/Spring Boot backends for enterprise ERP and Agri-tech systems, owning the complete workflow from UI to API to database.
- Engineered and documented RESTful APIs for inventory management, analytics, and operations modules; reduced bug backlog through systematic endpoint validation using Postman.
- Built Selenium-based automated regression test suites, increasing functional test coverage and reducing manual QA effort across release cycles.
- Containerized backend microservices with Docker, configured Kubernetes workloads, and supported production deployments on AWS (EC2, S3).
- Collaborated in Agile/Scrum cross-functional team using JIRA; consistently delivered production-ready code in two-week sprints.

Software Engineer Intern | *Ceylon Business Appliances (Pvt) Ltd* | Dehiwala, Sri Lanka **Jul 2022 – Jul 2023**

- Built a full-stack inventory management system from scratch — React frontend with Spring Boot backend — including RESTful APIs for product CRUD, stock tracking, and business workflows.
- Developed the Calmable Android mobile application in Java using Android Studio, focusing on UX design patterns that improved user engagement metrics.
- Collaborated with frontend and backend teams to integrate and test REST APIs across web and mobile platforms, supporting end-to-end QA and bug resolution.

PROJECTS

AI-Powered Banking Transaction Intelligence Platform — *GitHub: github.com/SamithaAthurupana/banking-intelligence-platforml*

- Architected a full-stack enterprise banking intelligence platform using FastAPI, PostgreSQL, React + TypeScript, and Docker, following clean architecture principles with strict separation of concerns across models, schemas, services, repositories, and API layers.
- Built a real-time fraud detection engine using Pandas for transaction pattern analysis and NumPy z-score anomaly detection; implemented a weighted risk scoring formula incorporating transaction amount, frequency, and geolocation anomalies to return risk levels (LOW / MEDIUM / HIGH).
- Designed RESTful API endpoints with proper HTTP status codes, Pydantic schema validation, and structured error handling; optimized PostgreSQL queries with composite indexes reducing transaction scan complexity.
- Developed a React + TypeScript frontend with a Dashboard Overview, Transaction Table, Fraud Alerts Panel, and Risk Score Charts (Recharts); containerized all services with Docker Compose and configured a CI/CD pipeline via Azure DevOps targeting Azure App Service and Azure PostgreSQL.

AI-Crypto-Intel — *GitHub: github.com/SamithaAthurupana/AI_Crypto_Intel*

- Built an async FastAPI backend with a full Retrieval-Augmented Generation (RAG) pipeline — covering ingestion, chunking, embeddings, FAISS vector search, and context injection — using LangChain and Pydantic.

- Developed a multi-step LangGraph autonomous agent to aggregate crypto news, perform sentiment analysis, query tenant knowledge bases, and generate automated reports.
- Designed multi-tenant vector store architecture with tenant isolation; deployed on AWS using Docker, Nginx, and GitHub Actions CI/CD pipelines with S3 storage integration.

Mind-Care App — *GitHub: github.com/SamithaAthurupana/mind_care_app*

- Built a cross-platform mental wellness mobile application in Flutter with conversational AI features and stress management tools.
- Developed a scalable FastAPI backend integrated with Hugging Face NLP models to deliver personalized guidance and NLP-driven mental wellness recommendations.
- Implemented adaptive UI layouts, state management, and modular code architecture enabling extensibility for mood tracking and guided exercises.

PharmaSys — *GitHub: github.com/SamithaAthurupana/PharmaSys*

- Built a full-stack pharmacy management application using JavaScript, CSS, Bootstrap, and FastAPI (Python), delivering a responsive UI with backend logic for managing products, employees, orders, and suppliers across complete business workflows.
- Integrated MySQL for persistent data storage and implemented Selenium-based automated UI test suites to validate application quality, reliability, and regression coverage across all core modules.
- Developed core CRUD workflows for inventory management, user roles, and business operations with structured RESTful API design, improving operational efficiency and data accuracy across pharmacy workflows.

LangChain Integration Project — *GitHub: github.com/SamithaAthurupana/langChain_integration*

- Built a backend integration system using Python, FastAPI, and LangChain to create LLM-powered workflows for chatbots, intelligent agents, and AI automation pipelines; designed modular service layers enabling reusable and extensible LLM orchestration across multiple use cases.
- Implemented LLM prompt management, chaining logic, and structured response handling with LangChain, enabling flexible multi-step reasoning workflows for chat interfaces and data processing pipelines with configurable chain composition.
- Developed a clean, modular backend architecture using FastAPI with clear separation of concerns across prompt templates, chain executors, and API routes; documented API endpoints and integration patterns to support team collaboration and future extensibility.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL

Frontend: React, Flutter, HTML5, CSS3, Android (Java)

Backend: FastAPI, Spring Boot, RESTful API Design, LangChain, MVC

Databases: MySQL, PostgreSQL, MongoDB, Firebase

Cloud & DevOps: AWS (EC2, S3), Docker, Kubernetes, GitHub Actions CI/CD, Nginx, Azure

AI / ML: LLM Orchestration, RAG Pipelines, LangGraph Agents, Hugging Face, FAISS, Prompt Engineering

Testing & Tools: Selenium, Postman, Git/GitHub, JIRA, Android Studio

Methodologies: Agile/Scrum, Full Stack Development, Test Automation, CI/CD

Certified : Azure certified: AZ-900, AI-900, AZ-900, AZ-104 | Postman Student Expert

EDUCATION

MSc in Information Technology (In Progress) | *SLIIT — Sri Lanka Institute of Information Technology* | Malabe | **Jun 2025 – Jun 2027**

BSc in Information Technology | *KDU — General Sir John Kotelawala Defense University* | Ratmalana | **Jan 2019 – Jan 2023**